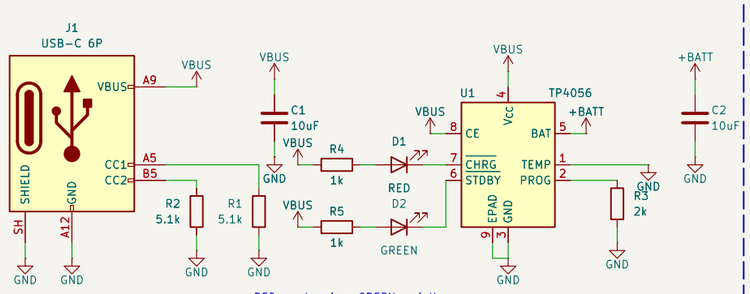
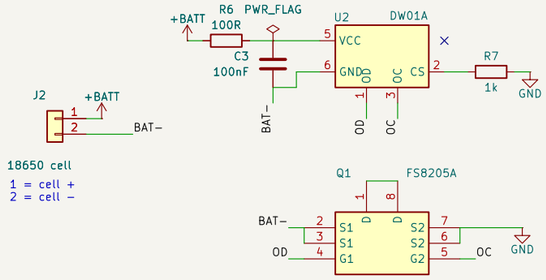
1. USB-C charging input (charge only, ~600 mA)



RED = charging, GREEN = full.
LEDs run from USB only (0 uA from the battery).

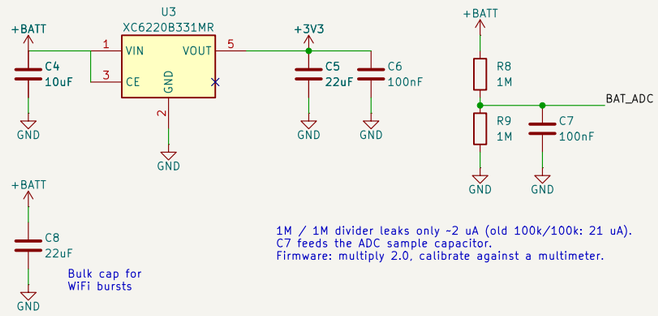
Charge current = 1200 V / R3 = 600 mA
TEMP tied to GND = no NTC.

2. Battery connector & protection (DW01A + FS8205A)



Cell(-) -> FS8205A -> board GND.
Cuts off at ~2.4 V, ~4.3 V, short circuit and ~3 A.
Quiescent current ~3 uA.

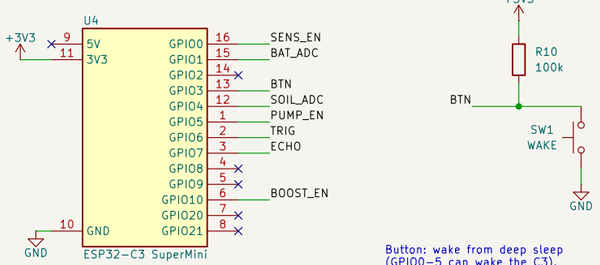
3. 3.3 V LDO (8 uA Iq) & battery measurement



1M / 1M divider leaks only ~2 uA (old 100k/100k: 21 uA).
C7 feeds the ADC sample capacitor.
Firmware: multiply 2.0, calibrate against a multimeter.

Bulk cap for
WiFi bursts

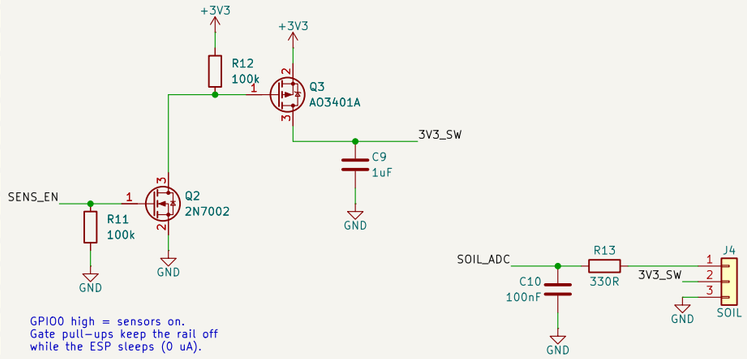
4. MCU — ESP32-C3 SuperMini (remove its red power LED!)



5V pin left open: the module is fed on 3V3 from U3,
bypassing its own LDO.
Flash over USB on the module or OTA.

Button: wake from deep sleep
(GPIO0-5 can wake the C3).
GPIO2/8/9 are strapping pins:
left unused on purpose.

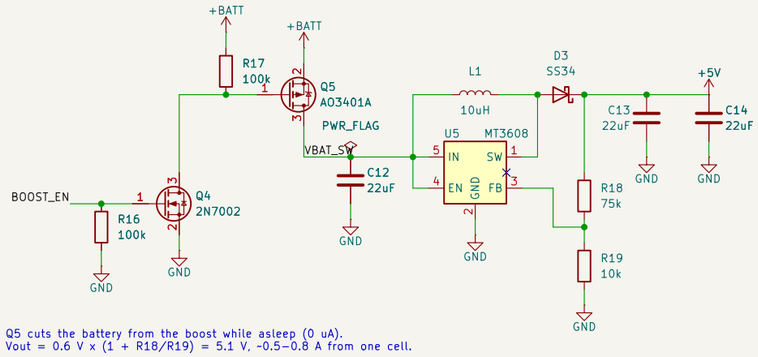
5. Switched 3.3 V sensor rail & soil moisture sensor



GPIO0 high = sensors on.
Gate pull-ups keep the rail off
while the ESP sleeps (0 uA).

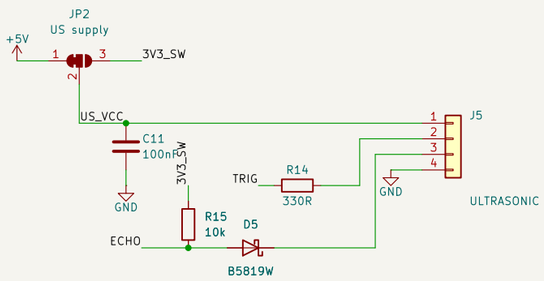
Capacitive Soil Moisture Sensor v2.0.0
J4: 1 = AOUT, 2 = VCC, 3 = GND

6. Switched 5 V boost (MT3608) for pump & ultrasonic



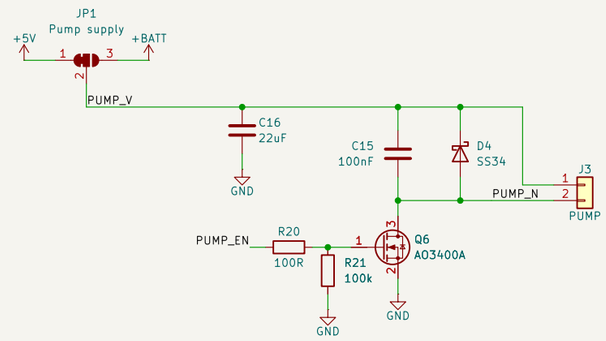
Q5 cuts the battery from the boost while asleep (0 uA).
Vout = 0.6 V x (1 + R18/R19) = 5.1 V, ~0.5-0.8 A from one cell.

7. Ultrasonic water level sensor (HC-SR04 / HC-SR04P)



JP2 default 1-2 = 5 V (works with HC-SR04 and HC-SR04P).
Cut and bridge 2-3 for a 3.3 V-only sensor.
ECHO goes through a Schottky + pull-up, so a 5 V echo
never reaches the ESP32-C3 pin.
J5: 1 = VCC, 2 = TRIG, 3 = ECHO, 4 = GND

8. Pump driver (submersible or peristaltic, up to 5 V)



JP1 default 1-2 = pump on the 5 V boost (5 V pumps).
Cut and bridge 2-3 to run a 3 V pump straight from the cell.
Firmware: switch BOOST_EN on before PUMP_EN.
D4 = flyback diode, C15 = motor noise.

Notes

GPIO map (same as the prototype + GPIO3, GPIO10):
GPIO0 SENS_EN GPIO1 BAT_ADC GPIO3 BTN GPIO4 SOIL_ADC
GPIO5 PUMP_EN GPIO6 TRIG GPIO7 ECHO GPIO10 BOOST_EN
Sleep budget: ESP32-C3 ~5 uA + XC6220 ~8 uA + DW01A ~3 uA + divider ~2 uA = ~18 uA
Check the SuperMini pin order against the silkscreen on the back of your module!



Generated with Claude Code — review before ordering
ESP32-C3 SuperMini, 18650 Li-ion, low-power design

Smart Plant Pot		
Sheet: /		
File: plantpot.kicad_sch		
Title: Smart Plant Pot — low power controller		
Size: A3	Date: 2026-10-13	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/1